

Dexter Fernandes

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ML Engineer

Machine Learning Engineer with 3+ years building and deploying computer vision systems for safety-critical clients (BP, Austin Industries), taking systems from initial concept through live deployment. I dig into why models fail (occlusion, lighting shifts, motion blur) using Grad-CAM, D-RISE, and TIDE error taxonomies, then close the gap with synthetic and adversarial training data. Applying that same playbook to biometric presentation-attack detection: a face liveness/anti-spoofing pipeline, tested for cross-dataset generalisation and audited for subgroup fairness. Skilled in Python, PyTorch, and TensorFlow.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Built benchmarking and profiling harnesses to evaluate models (YOLOv5, EfficientDet) across accuracy and latency post-quantisation alongside model-promotion gates to prevent quality regression.
- Designed adversarial-condition training and validation experiments (occlusion, lighting variation, motion artefacts) to benchmark robustness via F1 and failure-rate metrics, improving F1 by 15%.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.
- Built synthetic image generation and augmentation (OpenCV, scikit-image, albumentations) to cover edge cases and improve model robustness, reducing false positives and supporting long-term reliability.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.
- Debugged training dynamics and convergence issues in production CV models, balancing model complexity, augmentation, and validation strategy to improve generalisation under lighting variation, occlusion, motion artefacts, and camera contamination.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led design and deployment of a real-time collision-avoidance vision system for Austin Industries' construction sites. Set the detection criteria, built the dataset strategy, and owned evaluation from early prototypes to on-site rollout.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing robustness.
- Applied Grad-CAM and D-RISE for model interpretability and TIDE-based error taxonomy for failure analysis to diagnose recurring detection failures, prioritise dataset fixes, and reduce false positives by 33% across camera feeds.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Mentored junior engineers and collaborated with hardware, software, QA, and client teams to integrate perception modules and translate performance, latency, and failure analysis into rollout decisions for deployed safety systems.
- Presented model performance, failure analysis and deployment trade-offs to engineering, QA, hardware and client teams; findings directly shaped rollout and optimisation decisions.

PROJECTS

Presentation-Attack Detection under Generative AI

Jun 2026 - Present

- Building a face presentation-attack detection pipeline (photo, replay, and mask attacks), combining classical texture and frequency-domain features with a fine-tuned CNN, evaluated using APCER/BPCER/ACER/EER.
- Designing cross-dataset generalisation tests (training on CelebA-Spoof, evaluating on NUAA for print attacks; intra-dataset domain-shift protocol for replay/mask) alongside two augmentation strategies -- adversarial perturbation and procedurally-synthesised attack artefacts -- with Grad-CAM applied for failure-mode interpretability.
- Incorporating a subgroup-level fairness audit using a documented ITA skin-tone proxy, controlling for illumination and attack-medium confounds.
- Targeting ONNX/TensorRT edge deployment with latency profiling for real-time inference feasibility.

Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, YOLO26, RT-DETR, EfficientViT DINO, MobileSAM, and Depth Anything v2 to compare accuracy, latency, model size, and deployment suitability.

CERTIFICATIONS

Deep Learning Specialisation - FaceNet (Facial Verification coursework)

Jul 2019

deeplearning.ai

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed a real-time SLAM system on a Clearpath Husky using RTAB-Map, ROS2, and C++17, integrating LiDAR and radiological data for inspection-scenario localisation and mapping.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

SKILLS

- **Computer Vision & Machine Learning:** Convolutional Neural Networks, GRAD-CAM, NumPy, OpenCV, PyTorch, Scikit-Image, Scikit-Learn (sklearn), TensorFlow, TIDE (Toolbox for Identifying Detection Errors)
- **Edge Inference and Optimisation:** custom TensorRT plugins, FP16/INT8 deployment, latency/throughput profiling, Layer fusion, ONNX, OpenVINO, Optuna, Post-training Quantisation, Quantisation-Aware Training, Structured pruning
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, Docker, DVC (Data Version Control), FastAPI, Git, Health checks, Intel VPUs, Linux, Microsoft Azure, MLflow, model-promotion gates
- **Programming & Tooling:** Bash, C++17, CI/CD, Pandas, PostgreSQL, Python
- **Edge AI & Video Pipelines:** camera behaviour debugging, CUDA, FFmpeg, GStreamer, multi-stream video analytics, NVIDIA DeepStream 6.x/9.0, TensorRT
- **Biometrics & Liveness (In Progress):** Deep Learning Specialisation coursework), Presentation-attack detection (PAD) evaluation -- APCER/BPCER/ACER/EER, subgroup-level fairness auditing; facial verification fundamentals (FaceNet